

Improved lower bounds for the $3x + 1$ problem via the Krasikov–Lagarias inequalities at depths 3^{13} through 3^{20}

Martien de Jong*

July 2026 (draft)

Abstract

Let $\pi_1(x)$ denote the number of integers $n \leq x$ whose forward orbit under the $3x + 1$ map contains 1. Krasikov and Lagarias (2003) proved $\pi_1(x) \geq x^{0.84}$ for sufficiently large x by exhibiting feasible solutions of a linear-program family $L_k^{NT}(\lambda)$ at depth $k = 11$; this has remained the strongest unconditional lower bound (a December 2025 preprint, arXiv:2512.13760, still cites $x^{0.84}$ as the record). We solve $L_k^{NT}(\lambda)$ at depths $k = 13, 15, 17, 19, 20$ (up to 1,162,261,467 constraints) and verify explicit rational certificates in exact integer arithmetic, obtaining

$$\pi_1(x) \geq x^{0.9146} \quad \text{for all sufficiently large } x,$$

with intermediate verified exponents 0.8624, 0.8805, 0.8953, 0.9069. No new mathematics is required beyond [?]: Theorem 2.2 there converts any feasible solution into the density bound with exponent $\log_2 \lambda$. Our contribution is computational: a monotone (nonlinear Perron) iteration replaces the interior-point solver, making depths 3^{19} – 3^{20} feasible on a desktop machine, and every certificate is re-verified with all weights replaced by strict rational lower bounds. We further observe an exact scalar identity governing the family (the “min-loss identity”), which predicts the feasibility edges to three decimals and suggests the method attains $x^{1-\varepsilon}$ for every $\varepsilon > 0$.

1 The system

Fix $k \geq 2$ and let $[3^k] = \{m \bmod 3^k : m \equiv 2 \bmod 3\}$, $\alpha = \log_2 3$. Following [?, §2], the linear program $L_k^{NT}(\lambda)$ has principal variables $c^m \geq 1$ for $m \in [3^k]$, auxiliaries $\bar{c}^t = \min(c^t, c^{t+3^{k-1}}, c^{t+2 \cdot 3^{k-1}})$ for $t \in [3^{k-1}]$, and constraints

$$m \equiv 2 \pmod{9} : \quad c^m \leq c^{4m} \lambda^{-2} + \bar{c}^{(4m-2)/3} \lambda^{\alpha-2}, \tag{1}$$

$$m \equiv 5 \pmod{9} : \quad c^m \leq c^{4m} \lambda^{-2}, \tag{2}$$

$$m \equiv 8 \pmod{9} : \quad c^m \leq c^{4m} \lambda^{-2} + \bar{c}^{(2m-1)/3} \lambda^{\alpha-1}, \tag{3}$$

with indices taken mod 3^k resp. 3^{k-1} . By [?, Thm. 2.2], a feasible positive solution at parameter λ yields $\pi_a(x) \geq \Delta_1 c^m \lambda^y$ for every $a \not\equiv 0 \bmod 3$, hence a density exponent $\gamma = \log_2 \lambda$.

*With extensive machine assistance (Claude/Jengo); all results independently verifiable from the deposited certificates.

2 Method and results

The constraint map F_λ (right-hand sides, with auxiliaries substituted) is monotone, positively homogeneous and concave; feasibility is equivalent to its nonlinear Perron root being ≥ 1 , located by power iteration and bisection in λ . The implementation reproduces the published values at $k = 9$ ($\gamma = 0.8168$; cf. 0.81 in [?]) and $k = 11$ ($\gamma = 0.8418$; cf. 0.84 in [?]). At greater depths:

k	classes	λ_0 (rational)	min. slack	verified exponent
13	531,441	1818/1000	1.000279	0.8624
15	4,782,969	1841/1000	1.000307	0.8805
17	43,046,721	93/50	1.000504	0.8953
19	387,420,489	15/8	1.000939	0.9069
20	1,162,261,467	1885/1000	1.000194	0.9146

Theorem 1. $\pi_1(x) \geq x^{0.9146}$ for all sufficiently large x .

Verification. For each depth the deposited integer vector C (scaled Perron vector) is checked against all constraints with λ_0 rational and every weight replaced by a strict rational lower bound (80-digit decimal floors minus one ulp, common denominator 10^{18}); comparisons are exact integer arithmetic. Zero violations at all five depths (1.6×10^9 constraints in total). Certificates and a standalone `numpy`-only verifier are deposited with this note. Three independent cross-checks strengthen the claim: (i) a second, independently written implementation (different indexing and λ parametrization) reproduces the feasibility edges at overlapping depths to 3–4 decimals ($k=13$: 0.8620; $k=15$: 0.8801; $k=17$: 0.8950); (ii) the $k = 12$ certificate ($\gamma = 213/250 = 0.852$, already above the previous record) has been machine-verified in *Lean 4*: the coefficient lower bounds reduce to integer power inequalities (e.g. $p_a^{250} \leq 2^{250Q-426}$) and all 177,147 class inequalities are checked by compiled kernel evaluation, with no floating point anywhere; (iii) both implementations were calibrated against all published anchor values ($k = 2, 9, 11$).

3 The min-loss identity and the ceiling

Summing the edge equalities over all classes (the maps $m \mapsto 4m$ and the two branch maps are class bijections) collapses the system to a scalar law: at the Perron edge,

$$1 = \lambda^{-2} + \frac{q}{3} (\lambda^{\alpha-2} + \lambda^{\alpha-1}), \quad q = \frac{3 \sum_t \bar{c}^t}{\sum_m c^m}.$$

(Verified to 10^{-8} at $k = 5, \dots, 11$.) Note $q = 1$ forces $\lambda = 2$, i.e. $\gamma = 1$: the entire gap between this method and full density is the “min-loss” $1 - q$. Measured q rises 0.888 ($k=5$) $\rightarrow$ 0.9705 ($k=19$), driven by equalization of the refinement triples of the Perron vector; the within-vector heterogeneity profile decays geometrically in the trit position with bulk factor ≈ 0.84 , matching the subleading spectral ratio of the edge-linearized operator. If this contraction admits a uniform bound below 1 (supported by all measurements; we leave it as a conjecture), then $q \rightarrow 1$ and the method attains $\pi_1(x) \geq x^{1-\varepsilon}$ for every $\varepsilon > 0$, as hoped in [?].

Remark 1. Depth $k = 21$ (3.5×10^9 classes) is the natural next computation and decides between the extrapolations $\gamma_\infty \approx 0.976$ (endpoint Shanks) and $\gamma_\infty = 1$ (profile/contraction model); it requires roughly 64 GB of memory.

References

- [1] I. Krasikov, J. C. Lagarias, *Bounds for the $3x+1$ problem using difference inequalities*, Acta Arith. 109 (2003), 237–258; arXiv:math/0205002.
- [2] D. Applegate, J. C. Lagarias, *Density bounds for the $3x+1$ problem II: Krasikov inequalities*, Math. Comp. 64 (1995), 427–438.